

MICHAEL WU

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RESEARCH INTERESTS

My broad research interests are in the intersection of operating systems and computer architecture. I am specifically interested in the algorithms and heuristics that govern data movement in the presence of tiered memory and swapping.

EDUCATION

Yale University

PhD Computer Science (GPA: 4.0)

Advisor: Abhishek Bhattacharjee

New Haven, CT

Fall 2021 - Present

Rutgers University - New Brunswick

BS Computer Science and Mathematics (GPA: 3.95)

New Brunswick, NJ

Fall 2017 - Spring 2021

WORK EXPERIENCE

Google – Software Engineering Intern (PhD)

TPU Firmware Optimization (C++)

Sunnyvale, CA

Summer 2025

- Analyzed and mapped out complex driver-firmware interactions used to interface between host and accelerators for proprietary Tensor Processing Units (TPUs)
- Identified, profiled, and optimized latency sensitive operations contingent on costly host-to-accelerator communication

PUBLICATIONS

BULLETTIME: Time Dilation for High-Fidelity Tracing

International Symposium on Computer Architecture (ISCA 2026)

- *Authors:* **Michael Wu**, Sibren Isaacman, Anurag Khandelwal, Abhishek Bhattacharjee

Characterizing Emerging Page Replacement Policies for Memory-Intensive Applications

IEEE International Symposium on Workload Characterization (IISWC 2024)

- *Authors:* **Michael Wu**, Sibren Isaacman, Abhishek Bhattacharjee

Prefetching Using Principles of Hippocampal-Neocortical Interaction

Proceedings of the 19th Workshop on Hot Topics in Operating Systems (HOTOS 2023)

- *Authors:* **Michael Wu**, Ketaki Joshi, Andrew Sheinberg, Guilherme Cox, Anurag Khandelwal, Raghav Pothukuchi, Abhishek Bhattacharjee

Balancing Specialized Versus Flexible Computation in Brain-Computer Interfaces

IEEE Micro Top Picks from the 2020 Computer Architecture Conferences (Top Picks 2020)

- *Authors:* Ioannis Karageorgos, Karthik Sriram, Jan Vesely, Nick Lindsay, Xiayuan Wen, **Michael Wu**, Marc Powell, David Borton, Rajit Manohar, Abhishek Bhattacharjee

Hardware-Software Co-Design for Brain-Computer Interfaces

Proceedings of the 47th International Symposium on Computer Architecture (ISCA 2020)

- *Authors:* Ioannis Karageorgos, Karthik Sriram, Jan Vesely, **Michael Wu**, Marc Powell, David Borton, Rajit Manohar, Abhishek Bhattacharjee

RESEARCH

Yale University

Graduate Research Assistant

Prof. Abhishek Bhattacharjee, Prof. Anurag Khandelwal

2021-Present

- **Memory Oversubscription for LLM Inference (Under Submission):** Memory demands for LLMs scale far faster than accelerator on-chip memory, making offloading increasingly necessary, especially for Mixture-of-Experts models, whose total parameters are 5-10 $\times$ larger than dense models of comparable per-token cost. We develop an analytical framework for MoE memory oversubscription that derives closed-form throughput predictions for arbitrary caching and prefetching policies, parameterized by

expert activation probability, granularity, and hardware bandwidth. We show that offloading viability is governed by expert granularity, activation sparsity, and routing skew, with dense inference emerging as a limiting case. Our framework reveals that fine-grained sparse models can maintain over 80% throughput while offloading 10% of memory even over PCIe.

- **High-Fidelity Tracing:** Much computer architecture research relies on accurate traces for simulation, profiling, and debugging. However, traces can easily become inaccurate, as persisting traced data leads to bursty I/O overheads that can be asymmetric across threads. These overheads worsen as compute becomes more performant. This work formalizes the notion of accurate tracing as preserving the execution order of certain key actions in an application. We then use a "time dilation" approach to detect tracing overheads and inject delays when needed to preserve key ordering.
- **Characterizing and Optimizing MG-LRU for Tiered-Memory Systems:** Currently leading a project focused on developing adaptive techniques that would allow Linux's MG-LRU paging policy to dynamically adjust its behavior to best suit the current state of the system. This is based on our previous work showing that MG-LRU can induce significant performance variation under high memory pressure (see IISWC publication above).
- **Cognitively-Inspired Prefetching:** Examined the application of cognitive learning principles in the brain to develop efficient ML algorithms for prefetching memory pages in swap and remote-memory contexts.

Yale University

Undergraduate Research Assistant

Prof. Abhishek Bhattacharjee

2019

- **HALO:** HALO is a general-purpose architecture designed for implantable brain-computer interfaces. My work focused on using high-level synthesis techniques to develop low-power hardware implementations of signal processing algorithms (e.g. fast fourier transform, cross-correlation, discrete wavelet transform).

TEACHING

- TA for CPSC 420: Computer Architecture (Yale University) *Spring 2022*
- TA for CPSC 323: Systems Programming and Computer Organization (Yale University) *Fall 2022*
- TA for CS 416: Operating Systems at (Rutgers University) *Spring 2020*

AWARDS AND HONORS

- **National Science Foundation Graduate Research Fellow (NSF GRFP)** *2022*
- **Summa Cum Laude Graduate - Rutgers University** *2022*
- **Presidential Scholarship - Rutgers University** *2017 - 2022*